Francisco Madaleno

PhD Fellow @ MLSM DTU

Education

Technical University of Denmark (DTU)

PhD Candidate @ **Machine Learning for Smart Mobility (MLSM)**

October 2024 - Present, Copenhagen, Denmark

PhD Topic: Active Learning Causal Abstractions on Simulators for Climate Change Policy Making

Instituto Superior Técnico / MSc Aerospace Engineering

2014 - 2019, Lisbon, Portugal, GPA: 16/20

MSc Thesis: "Flying Tourist Problem: An Integer Linear Programming Approach" as a Research Student at **INESC-ID, Lisbon**. An optimization tool to find the best schedule and route for a multi-city flight trip with real-time data integrated in a web application. Mark: 18/20

Athens Programme 2019: Operations Research in the Airline Industry, **Mines Paristech**
Ten-day intensive course with **AirFrance KLM** Operations Research/Data Science Team.

Summer Course 2018: Experimental Aerodynamics, **University of Southern Denmark**
Optimization project of a wind turbine blade for **LM Wind Power**.

Polytechnic University of Catalonia, UPC - ESEIAAT / Erasmus

2017 - 2018, Barcelona, Spain

Erasmus exchange program: Specialization in Optimization and Operations Research.

Experience

Zalando SE / Applied Scientist

September 2021 - August 2024, Berlin, Germany

Logistics and Algorithms. Developed strategies and algorithms for logistics networks to solve assignment and flow problems. Built and managed micro-service infrastructure to run highly-parallelizable algorithms. Contributed to Zalando's open- and inner-source projects as well as to the company's data science community.
Presented in GOR Workshop 2022: Applied Methods from Mathematical Optimization and Machine Learning for e-commerce. "Mathematical Optimization Meets Machine Learning to Optimize Stock Distribution at Zalando"

Zalando SE / Intern Applied Scientist

April 2021 - September 2021, Berlin, Germany

Pricing and forecasting. Collaborated on a MIP tool that consumes a forecast of over 500k items across 14 markets to recommend discounts over the period of 26 weeks.
Developed a heuristic that defines feasible target ranges as bounds for the MIP tool.

von Karman Institute / INOV Contacto 2020

February 2020 - September 2020, Brussels, Belgium

Participated in the program INOV Contacto 2020 funded by AICEP Portugal, in which I worked for the **von Karman Institute for Fluid Dynamics** as a Research Intern Engineer.
At von Karman, I was responsible for building a computational model and simulation tool to optimize a system of a turbine for the **Safran Group.**

Volunteering

ReDI School of Digital Integration

October 2023 - October 2024, Berlin, Germany

Teacher Assistant in Introduction to Computer Science. Taught Python fundamentals and assisted students in their first semester final project.
ReDI is a non-profit tech school providing migrants and marginalized locals free and equitable access to digital education.

Publications

Peer-Reviewed:

- Coarsening Causal DAG Models, (Accepted for Oral @ CLeaR 2026, published in PMLR), 2026
- Bayesian Hierarchical Invariant Prediction, (Accepted for Poster @ CLeaR 2026, published in PMLR), 2026

Preprints:

- Coarsening Linear Non-Gaussian Causal Models with Cycles, 2026

Oral Presentations

2026

- Contributed Talk at the 5th Conference on Causal Learning and Reasoning (CLeaR) 2026: Coarsening Causal DAG Models

2025

- Contributed Talk at 41st Conference on Uncertainty in Artificial Intelligence Workshop on Causal Abstractions and Representations: Bayesian Hierarchical Invariant Prediction

Conferences Attended

2026

- 5th Conference on Causal Learning and Reasoning (CLeaR) 2026

2025

- 4th CLeaR (Causal Learning and Reasoning) 2025
- 41st Conference on Uncertainty in Artificial Intelligence
- Fast Machine Learning for Science Conference 2025
- EurIPS 2025

Summer Schools

2026

- Oxford Machine Learning Summer School: MLx Representation Learning & Gen AI

2025

- Cambridge Ellis Unit Summer School on Probabilistic Machine Learning 2025

Poster

2026

- 5th Conference on Causal Learning and Reasoning (CLeaR) 2026: Coarsening Causal DAG Models
- 5th Conference on Causal Learning and Reasoning (CLeaR) 2026: Bayesian Hierarchical Invariant Prediction

2025

- Cambridge Ellis Unit Summer School on Probabilistic Machine Learning 2025: Bayesian Hierarchical Invariant Prediction
- 41st Conference on Uncertainty in Artificial Intelligence @ Workshop on Causal Abstractions and Representations: Bayesian Hierarchical Invariant Prediction
- AI in Science Summit (AIS2025): The APEX project - Artificial Intelligence for Policy Excellence in the Climate Crisis: initial results and outlook
- EurIPS 2025 @ Causality for Impact Workshop: Case Study: Learning Interventionally Consistent Surrogates for Climate Adaptation

2022

- Improving Zalando's Warehouse Efficiency by Relocating Items, International Conference on Operations Research, Karlsruhe, Germany, 2022

Teaching

2026:

- Model-based Machine Learning

Teaching Assistant

2026:

- Model-based Machine Learning
- Advanced Business Analytics

Reviewing

2026:

- The Fortieth Annual Conference on Neural Information Processing Systems NeurIPS 2026),
- Journal of the Royal Statistical Society, Series B,

- npj Urban Sustainability

2025:

- Journal of the Royal Statistical Society, Series B,
- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)

Grants / Funding

2026

- Thomas B. Thriges Fond for Research Visit

Supervision

- Filip Arthur Blaafjell (MSc Thesis)

Community Engagement

2026

- Organized seminar on Spatio-Temporal Machine Learning (Invited Speaker: Gerrit Grossmann (DFKI))
- Presented Coarsening Causal DAG Models at CoCaLa group at the University of Copenhagen
- Participated and organised meetings of the CPHDisco Network (Coordinated by Anne Helby Petersen from the University of Copenhagen)

2025

- Organised seminar on Causal Representation Learning (Invited Speaker: Simon Bing (TU Berlin))
- Participated and organised meetings of the CPHDisco Network (Coordinated by Anne Helby Petersen from the University of Copenhagen)